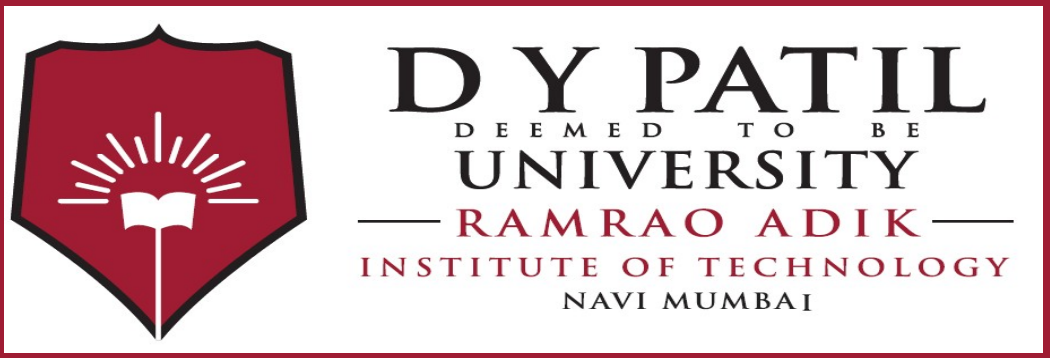
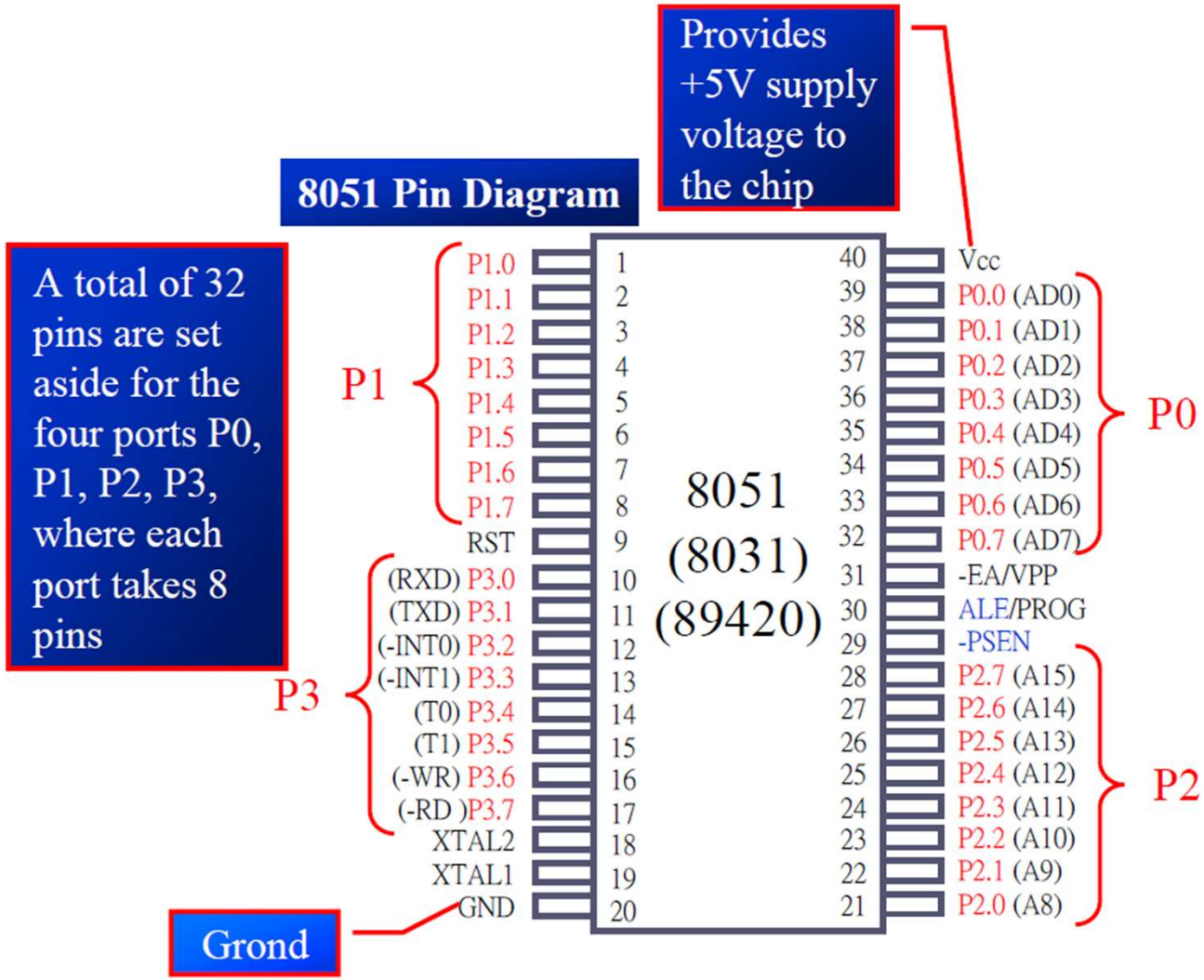


Embedded System
(Multidisciplinary Minor)
Module 5: 8051 Peripheral Interfacing

8051 Microcontroller-Pin Diagram



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LCD Interfacing

LCD INTERFACING

LCD Operation

- LCD is finding widespread use replacing LEDs
  - The declining prices of LCD
  - The ability to display numbers, characters, and graphics
  - Incorporation of a refreshing controller into the LCD, thereby relieving the CPU of the task of refreshing the LCD
  - Ease of programming for characters and graphics

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LCD INTERFACING

LCD Pin Descriptions

Pin Descriptions for LCD

Pin	Symbol	I/O	Descriptions
1	VSS	--	Ground
2	VCC	--	+5V power supply
3	VEE	--	Power supply to control contrast
4	RS	I	RS=0 to select command register, RS=1 to select data register
5	R/W	I	R/W=0 for write, R/W=1 for read
6	E	I/O	Enable
7	DB0	I/O	The 8-bit data bus
8	DB1	I/O	The 8-bit data bus
9	DB2	I/O	The 8-bit data bus
10	DB3	I/O	The 8-bit data bus
11	DB4	I/O	The 8-bit data bus
12	DB5	I/O	The 8-bit data bus
13	DB6	I/O	The 8-bit data bus
14	DB7	I/O	The 8-bit data bus

used by the LCD to latch information presented to its data bus

- Send displayed information or instruction command codes to the LCD
- Read the contents of the LCD's internal registers

LCD INTERFACING

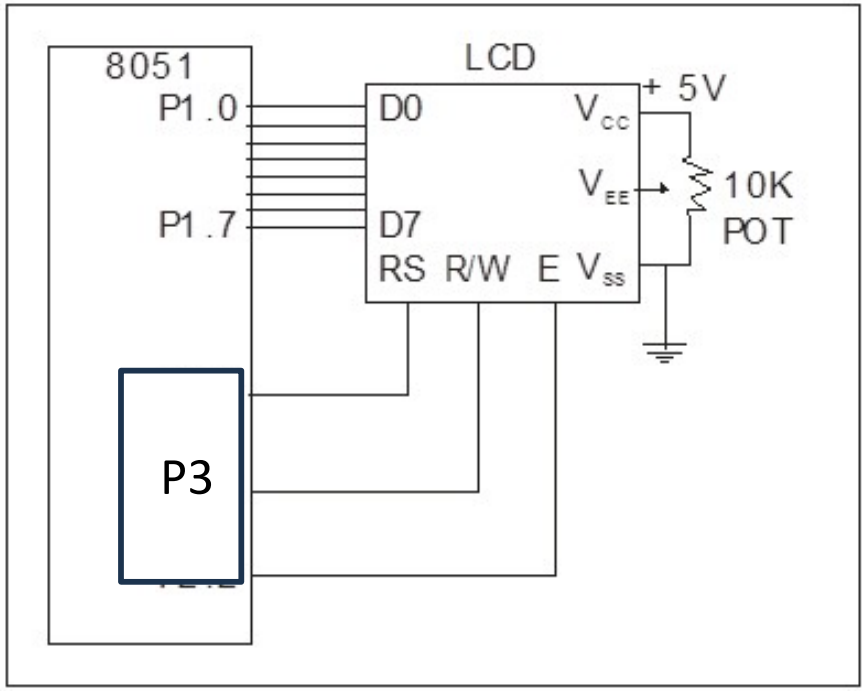
LCD Command Codes

LCD Command Codes

Code (Hex)	Command to LCD Instruction Register
1	Clear display screen
2	Return home
4	Decrement cursor (shift cursor to left)
6	Increment cursor (shift cursor to right)
5	Shift display right
7	Shift display left
8	Display off, cursor off
A	Display off, cursor on
C	Display on, cursor off
E	Display on, cursor blinking
F	Display on, cursor blinking
10	Shift cursor position to left
14	Shift cursor position to right
18	Shift the entire display to the left
1C	Shift the entire display to the right
80	Force cursor to beginning to 1st line
C0	Force cursor to beginning to 2nd line
38	2 lines and 5x7 matrix

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LCD 8051 Interfacing



LCD Pin	Function	Connection in the given code
RS	Register Select	`P3.5` (Instruction/Data selection)
RW	Read/Write	`P3.4` (Always write: 0)
EN	Enable	`P3.3` (Latch data to LCD)
D0-D7	Data Lines (8-bit)	`P1` (Port 1 sends data/commands)

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Register Select and Read/Write

RS (Register Select)
Function:
Selects the **register** to be used:

- `RS = 0`: Instruction Register (Command Mode)
- `RS = 1`: Data Register (Character/Data Mode)

Connection in Code:
Connected to **`P3.5`**

- `CLR P3.5` → Command mode
- `SETB P3.5` → Data mode

RW (Read/Write)
Function:
Selects **Read or Write operation**:

- RW = 0: Write to LCD
- RW = 1: Read from LCD (usually not used unless checking busy flag)

Connection in Code:
Connected to **P3.4**

- The code always uses CLR P3.4 → **Write mode only**

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Enable and Data Bus Lines

EN (Enable)
Function:
Used to **latch (lock in) data/command** into the LCD.
Data is processed by the LCD **on the falling edge** of the EN pulse.
Connection in Code:
Connected to **P3.3**

SETB P3.3 → EN = 1
CLR P3.3 → EN = 0

A quick high-to-low pulse (EN=1 → EN=0) tells the LCD to read the inputs.

D0–D7 (Data Bus Lines)
Function:
These are the **8 data lines** used to send:

- **Commands** (e.g., clear screen, set cursor)
- **Data** (characters to display)

Connection in Code:
Connected to **Port 1 (P1)**

```
MOV P1,A   ; A holds the command or character
```


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Code Execution Flow

1. **Put the value** (command or ASCII) in **register A**
2. **Move it to P1** (which is connected to LCD D0–D7)
3. **Set RS:**
    1. RS = 0 for command
    2. RS = 1 for data
4. **RW = 0** (always write)
5. **Pulse EN = 1 → 0** to latch the value into the LCD

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KEYBOARD INTERFACING

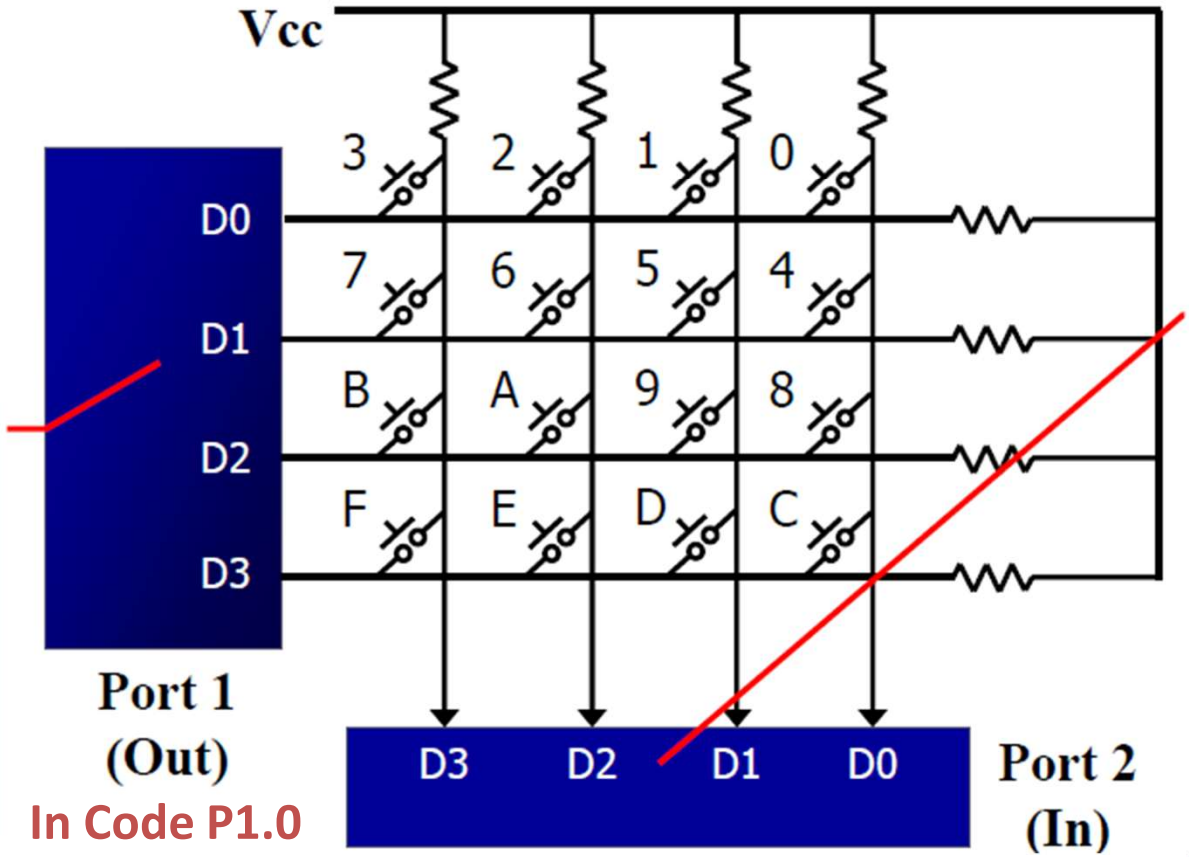
- Keyboards are organized in a matrix of rows and columns
  - The CPU accesses both rows and columns through ports
    - Therefore, with two 8-bit ports, an 8 x 8 matrix of keys can be connected to a microprocessor
  - When a key is pressed, a row and a column make a contact
    - Otherwise, there is no connection between rows and columns
- In IBM PC keyboards, a single microcontroller takes care of hardware and software interfacing

KEYBOARD INTERFACING

Scanning and Identifying the Key

- A 4x4 matrix connected to two ports
  - The rows are connected to an output port and the columns are connected to an input port

Matrix Keyboard Connection to ports



If all the rows are grounded and a key is pressed, one of the columns will have 0 since the key pressed provides the path to ground

If no key has been pressed, reading the input port will yield 1s for all columns since they are all connected to high (V_{cc})

KEYBOARD INTERFACING

Grounding Rows and Reading Columns

- It is the function of the microcontroller to scan the keyboard continuously to detect and identify the key pressed
- To detect a pressed key, the microcontroller grounds all rows by providing 0 to the output latch, then it reads the columns
  - If the data read from columns is D3 – D0 = 1111, no key has been pressed and the process continues till key press is detected
  - If one of the column bits has a zero, this means that a key press has occurred
    - For example, if D3 – D0 = 1101, this means that a key in the D1 column has been pressed
    - After detecting a key press, microcontroller will go through the process of identifying the key

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KEYBOARD INTERFACING

Grounding Rows and Reading Columns (cont')

- Starting with the top row, the microcontroller grounds it by providing a low to row D0 only
  - It reads the columns, if the data read is all 1s, no key in that row is activated and the process is moved to the next row
- It grounds the next row, reads the columns, and checks for any zero
  - This process continues until the row is identified
- After identification of the row in which the key has been pressed
  - Find out which column the pressed key belongs to

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Go to Code

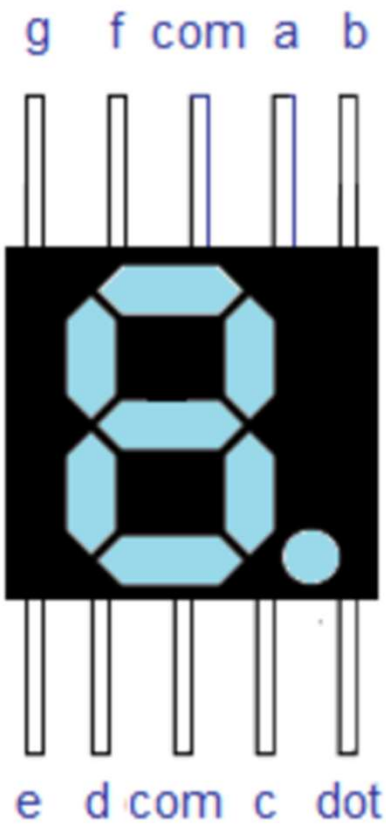
START
↓
Set DPTR to LUT
Initialize P0
↓
SCAN keypad row-by-row
└ For each row, check columns
└ If key press detected:
└ Load hex value
└ Call DISPLAY
└ Fetch pattern from LUT
└ Output to P0
GO TO START

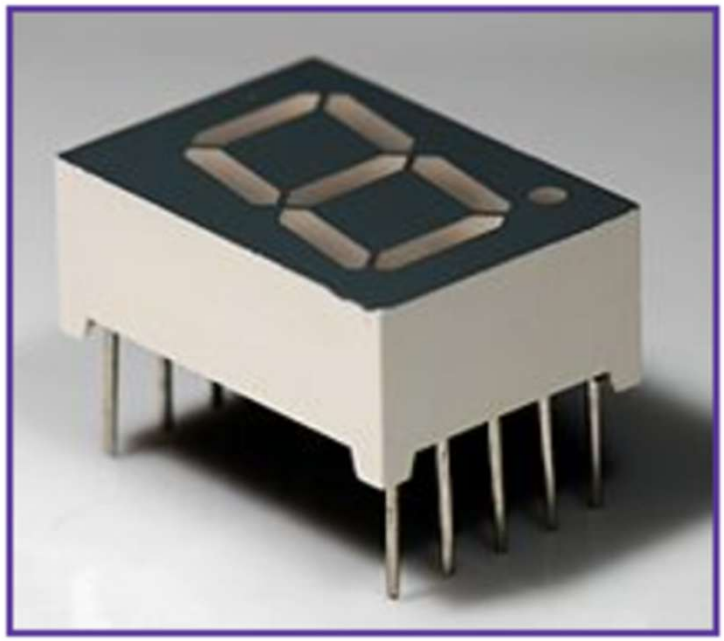
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Code for Keyboard

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8051 and Seven Segment Display Interfacing





7 segment LED display

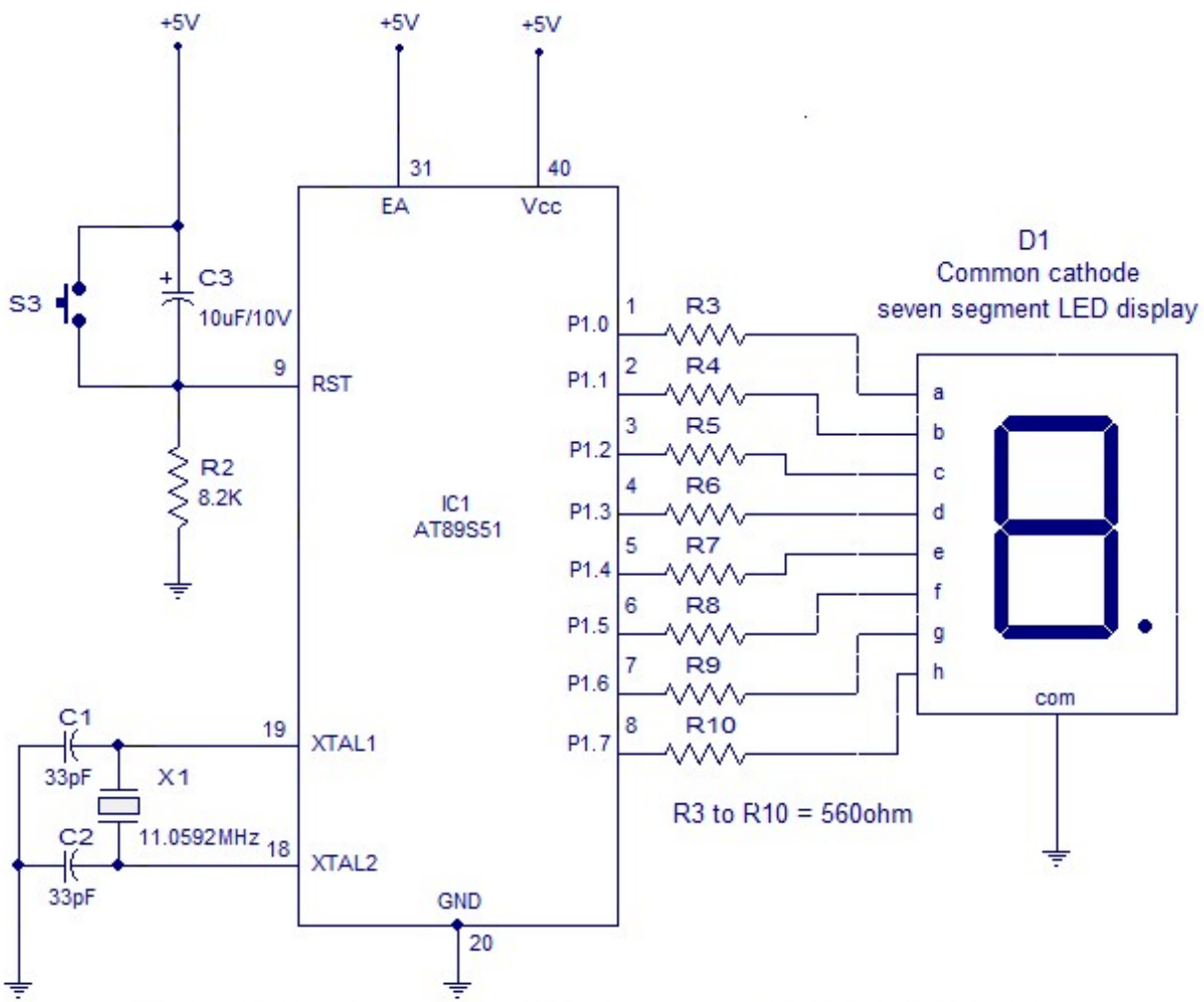
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8051 and Seven Segment Display Interfacing

- 7 segment LED display is very popular and it can display digits from 0 to 9 and quite a few characters like A, b, C, ., H, E, e, F, n, o,t,u,y, etc.
- Knowledge about how to interface a seven segment display to a micro controller is very essential in designing embedded systems.
- A seven segment display consists of seven LEDs arranged in the form of a squarish **‘8’** slightly inclined to the right and a single LED as the dot character.
- Different characters can be displayed by selectively glowing the required LED segments. S

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8051 and Seven Segment Display Interfacing



Interfacing 7 segment display to 8051 www.circuitstoday.com

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8051 and Stepper Motor Interfacing

A Stepper Motor is widely used device that translates electrical pulses into mechanical movements.

Applications: Disk Drives, Dot Matrix Printer, and Robotics, Stepper Motor is used for Position Control.

Stepper Motor have: Rotor surrounded by a stators

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Stepper Motor

Full Step Mode where two coils are energized once. Other modes: Wave Drive Mode, Half Step Mode

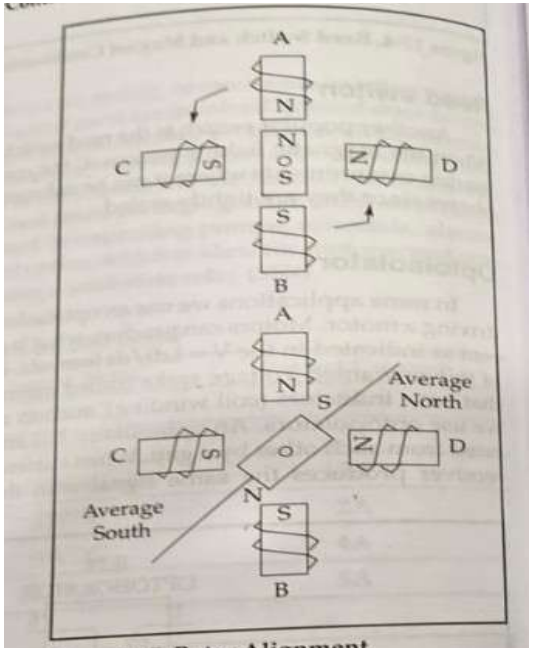


Figure 17-7. Rotor Alignment

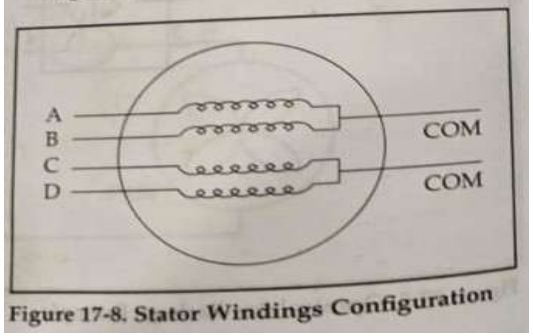


Figure 17-8. Stator Windings Configuration

Table 17-3: Normal 4-Step Sequence

Clockwise	Step #	Winding A	Winding B	Winding C	Winding D	Counter-Clockwise
	1	1	0	0	1	
	2	1	1	0	0	
	3	0	1	1	0	
	4	0	0	1	1	

Table 17-4: Stepper Motor Step Angles

Step Angle	Steps per Revolution
0.72	500
1.8	200
2.0	180
2.5	144
5.0	72
7.5	48
15	24

Stepper Motor and 8051 Interfacing

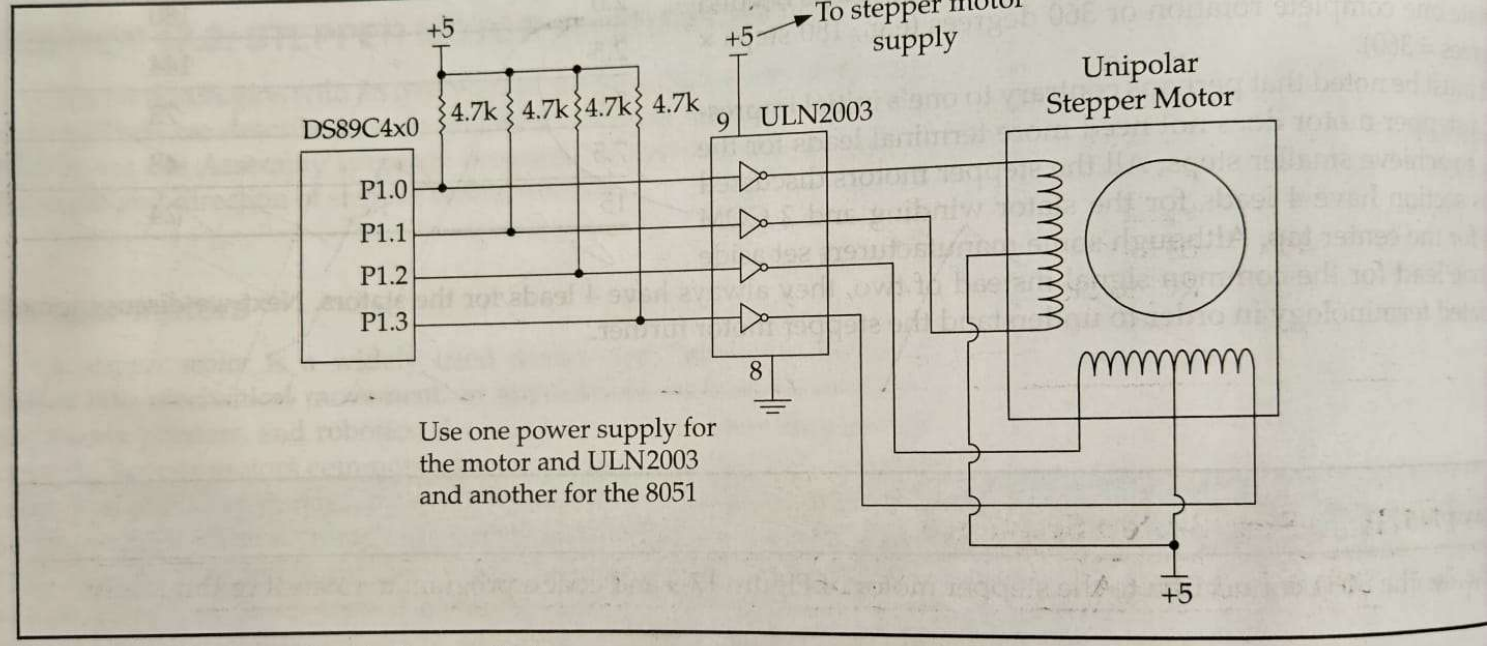


Figure 17-9. 8051 Connection to Stepper Motor

Steps per second and rpm relation

The relation between rpm (revolutions per minute), steps per revolution, and steps per second is as follows.

$$\text{Steps per second} = \frac{rpm \times \text{Steps per revolution}}{60}$$

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8051 Code for Stepper Motor Interfacing

```
                MOV A,#0CCH          ; full-step sequence, the first entry is 1100=C.
                                     ; here CC is used for continuous rotation.
REPEAT:         MOV P2,A    ; send sequence to motor.
                RR A                 ; next step of the sequence to rotate in clockwise direction.
                LCALL DELAY; wait before going to the next step.
                SJMP REPEAT          ; next step.

DELAY:          MOV R3,#01; delay, it can be varied to change the speed of operation.
L3:             MOV R1,#50; here we have used delay program written using simple registers.
L2:             MOV R2,#0FFH         ; The count of R1, R2, R3 registers decide the time of delay.
L1:             DJNZ R2,L1   ; This delay program represents simplest type of delay logic.
                DJNZ R1,L2
                DJNZ R3,L3
                RET

                END
```


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DC Motor Interfacing

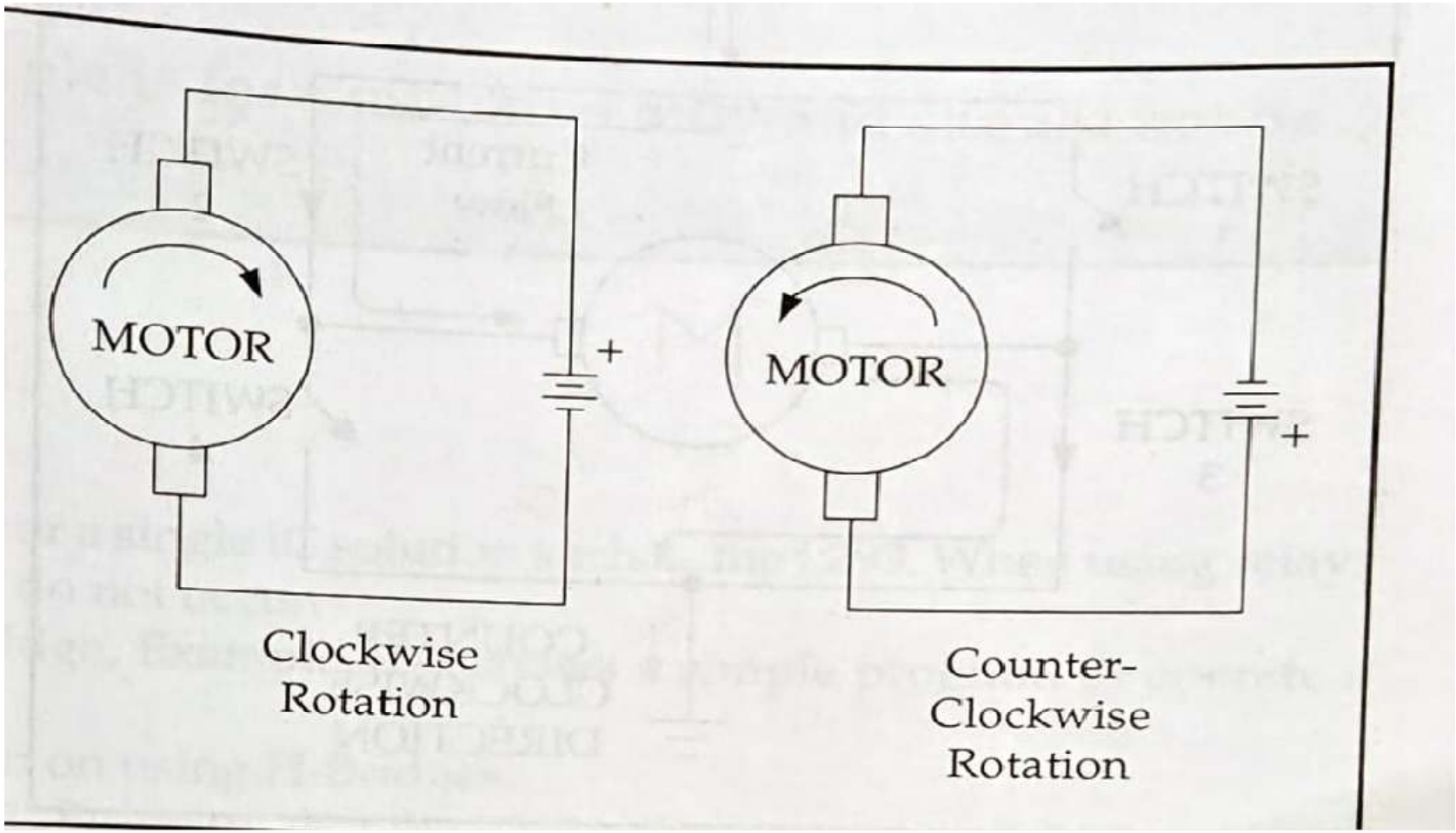
DC motor is another widely used device that translates electrical pulses into mechanical movements. In DC Motor we have + and – Leads.

Connecting them to a DC voltage source moves the motor into one direction. By reversing the polarity, DC motor will move in opposite direction.

Maximum speed of DC motor is provided in its datasheet.

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DC Motor Interfacing



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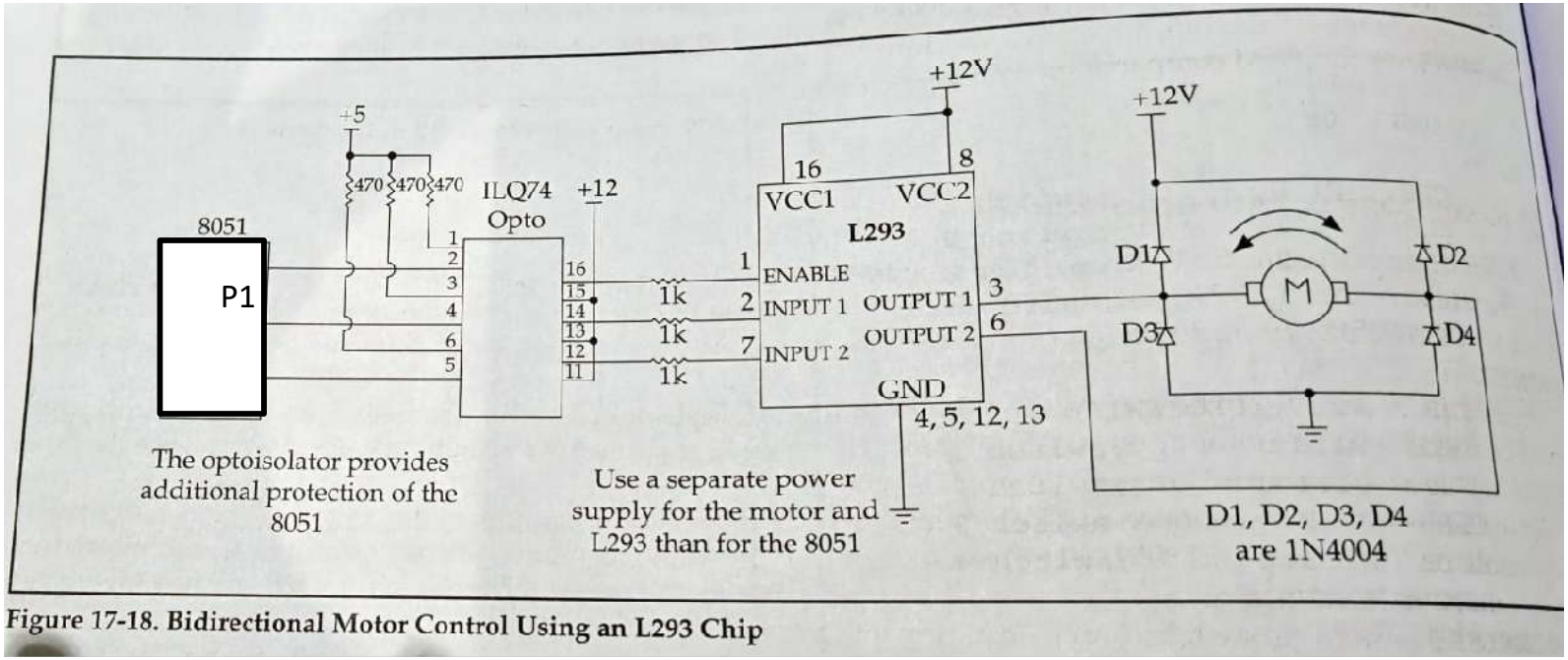


Figure 17-18. Bidirectional Motor Control Using an L293 Chip

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Motor Configurations

Table 17-10 shows some of the logic configurations for the H-Bridge design.

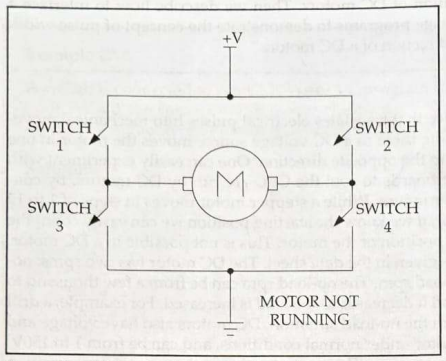


Figure 17-14. H-Bridge Motor Configuration

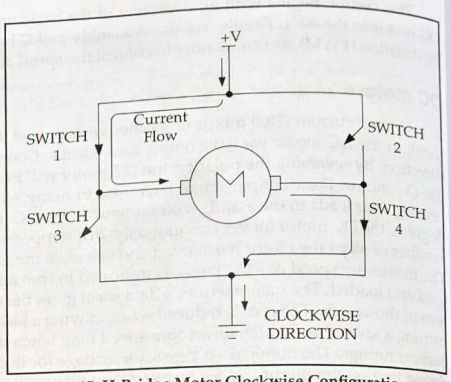


Figure 17-15. H-Bridge Motor Clockwise Configuration

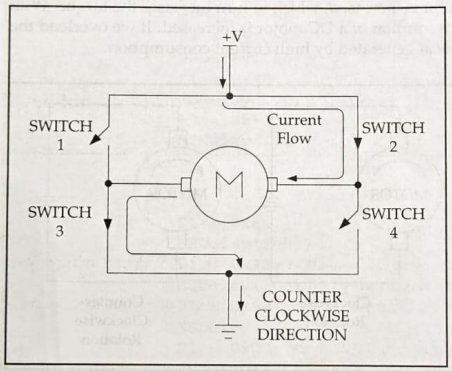


Figure 17-16. H-Bridge Motor Counterclockwise Configuration

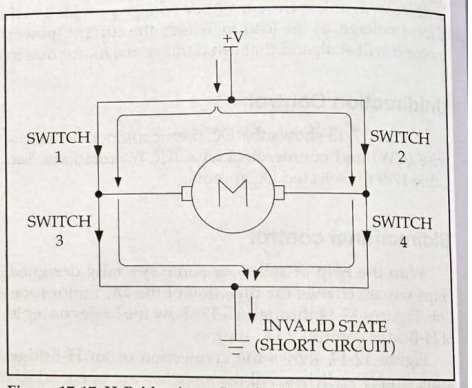


Figure 17-17. H-Bridge in an Invalid Configuration

Table 17-10: Some H-Bridge Logic Configurations for Figure 17-14

Motor Operation	SW1	SW2	SW3	SW4
Off	Open	Open	Open	Open
Clockwise	Closed	Open	Open	Closed
Counter Clockwise	Open	Closed	Closed	Open
Invalid	Closed	Closed	Closed	Closed

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Thank You

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